



BUSINESS ANALYSIS

Lecture 3: Four Types of Business Analysis

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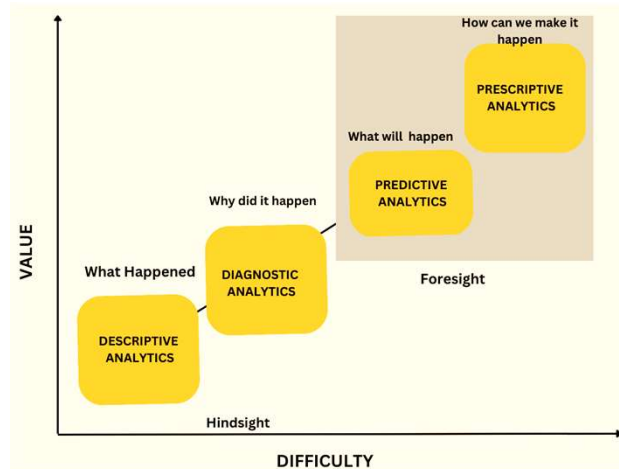


WHAT WE HAVE LEARNT SO FAR

- General principles all businesses should focus on to optimize profits, output and service quality:
 - Process efficiency
 - Reducing defects
 - Maximizing customer/client satisfaction
- Business frameworks such as TQM, Six Sigma, ISO 9001 aim at enshrining these principles
- The relationship between business analytics and business methodologies
 - Methodologies provide the framework for how business work is organized
 - Analytics provides insights needed to make the methodologies effective
- Examples of the relationship between business frameworks and business analytics:
 - Analysis of data to reducing defects
 - Analysis of process efficiency to increase customer satisfaction
- *Having understood the importance of business frameworks and analytics, what should a business do next?*

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FOUR TYPES OF BUSINESS ANALYTICS



Source: Alma Better Bytes; <https://www.almabetter.com/bytes/articles/four-types-of-data-analysis>
 Harvard Business School Online: <https://online.hbs.edu/blog/post/types-of-data-analysis>

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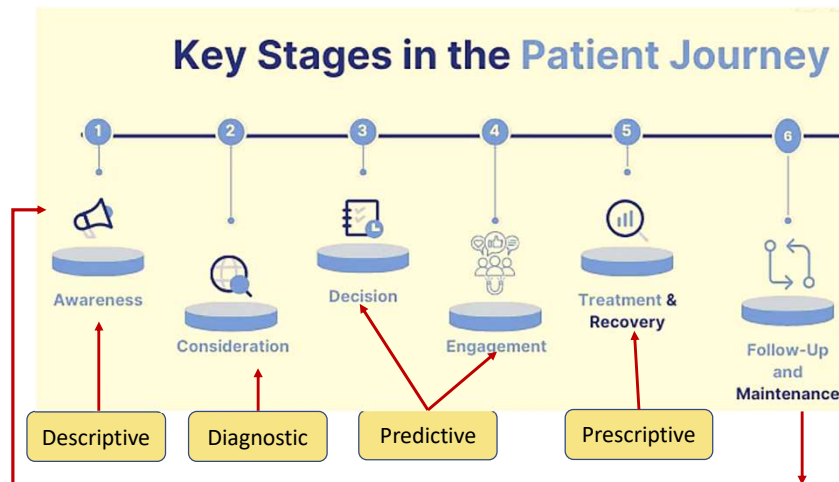
FOUR TYPES OF BUSINESS ANALYTICS: AN OVERVIEW

- Video "What is Business Analytics" <https://online.hbs.edu/blog/post/types-of-data-analysis>
- **The Four Types of Business Analytics**
 - **Descriptive Analytics**
 "Descriptive analytics is the simplest type of analytics and the foundation the other types are built on. It allows you to pull trends from raw data and succinctly describe what happened or is currently happening.
 Descriptive analytics answers the question, "What happened?"
 - **Diagnostic Analytics**
 "Diagnostic analytics addresses the next logical question, "Why did this happen?"
 Taking the analysis a step further, this type includes comparing coexisting trends or movement, uncovering correlations between variables, and determining causal relationships where possible."
 - **Predictive Analytics**
 "Predictive analytics is used to make predictions about future trends or events and answers the question, "What might happen in the future?"
 By analyzing historical data in tandem with industry trends, you can make informed predictions about what the future could hold for your company."
 - **Prescriptive Analytics**
 "Finally, prescriptive analytics answers the question, "What should we do next?"
 Prescriptive analytics takes into account all possible factors in a scenario and suggests actionable takeaways. This type of analytics can be especially useful when making data-driven decisions."

Source: Harvard Business School Online: <https://online.hbs.edu/blog/post/types-of-data-analysis>

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TYPES OF BUSINESS ANALYTICS: SOME SIMILARITIES WITH VISIT TO A DOCTOR



Source: WaveTech; <https://www.wavetec.com/patient-experience/patient-journey-mapping/>

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EXAMPLES OF THE FOUR TYPES OF BUSINESS ANALYTICS

Descriptive Analytics (*What happened?*)

Imagine you are a manager in a company X running a retail store of electronic products. As a manager it is your duty to document month-wise sales of product Y. You look at the data of month-wise expenses of the company from January to June and notice that whereas expenses are rising, sales are decreasing.

Diagnostic Analytics (*Why did it happen?*)

You analyze the head-wise expenses and notice that it is the electricity expenses that have increased from January to June. Next, you analyze the product-wise sales and conclude that the dip in sales has occurred mainly due to fewer sales of heating appliances.

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EXAMPLES OF THE FOUR TYPES OF BUSINESS ANALYTICS

Predictive Analytics (*What may happen in the future?*)

The customers at the store are mainly buying products with seasonal utilities. Therefore, total sales depend on whether the store is offering seasonal products.

Prescriptive Analytics (*What should we do?*)

Change of inventory (stock of goods)

- The store needs to diversify its range of seasonal products and also offer products that are not season dependent (e.g., stove tops)

Change marketing strategy

- Focus of marketing should change with seasons with strategies such as discount offers on heating appliances in November and cooling appliances in March
- The store should intensify its marketing of non-seasonal products

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TEN MINUTES ACTIVITY

- Descriptive Analytics
 - Can you draw a table of some hypothetical business data?
 - Can you draw a hypothetical chart of this data?
 - What initial information/conclusion can you derive from this data?
- Diagnostic Analytics
 - What do you think are the reasons for the trend depicted in the table and the chart?
 - Can you draw a hypothesis to encapsulate the reason explaining the trend?

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ANALYTICS AT WORK

The case of Costco (a US store for bulk buying)

- Costco in the US is known for offering a wide selection of items at relatively low prices, but the club's \$65 annual membership fee can deter some shoppers
- Savings from occasional discounts on big-ticket items such as TVs, mattresses, and household appliances can easily exceed the cost of a year's membership, but those purchases are expensive and infrequent for most household
- **Question: Which product categories do you think would justify a Costco membership the \$65 annual fee?**

Hypothesis:

Can you draft a hypothesis for the above question?

Source: MSN; *Business Insider*; Costco shoppers say these items basically make a membership pay for itself. We did the math; <https://www.msn.com/en-us/money/companies/costco-shoppers-say-these-items-basically-make-a-membership-pay-for-itself-we-did-the-math/ss-AA1TzJA7?ocid=hpmnsn&cvid=69736122134a4db9a1275a7924269190&ei=63>

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ANALYTICS AT WORK

The case of Costco: Method for analytics used by researchers

- Product identification
 - An initial list of products was drawn from the most popular responses in the Costco subreddit
 - The list was downsized to products (e.g., product Z) that are consumed steadily throughout the year
 - The prices at Costco for these products were compared to prices at other retailers, such as Walmart, Amazon, Target
- Product consumption data
 - ChatGPT was consulted to estimate how much of each item a typical two-person household would use in a year (e.g., $Z * 100$)
 - Annual costs were worked out for the household based on those quantities (e.g., $100 Z * \text{Price } \$10 \text{ per } Z = \$1,000$)
 - Price \$10 per Z was compared with national brands available in many retailers for comparison (product Z was cheaper at Costco!)
- Math
 - A payback score was allocated for each item to represent
 - Annual savings at Costco for Product Z vis-à-vis other retailers
 - How much of the \$65 membership fee is covered in these savings
 - E.g., If payback score is 100, savings = fee; if payback score is 200, savings = 2 fee; if payback score is 50 savings = 1/2 fee

MSN; *Business Insider*; Costco shoppers say these items basically make a membership pay for itself. We did the math; <https://www.msn.com/en-us/money/companies/costco-shoppers-say-these-items-basically-make-a-membership-pay-for-itself-we-did-the-math/ss-AA1TzJA7?ocid=hpmnsn&cvid=69736122134a4db9a1275a7924269190&ei=63>
 New York Post: <https://nypost.com/2026/01/13/lifestyle/costco-items-that-make-the-65-annual-membership-worthwhile/>

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TEN MINUTES ACTIVITY

- What do you think was the formula for calculating the Payback Score?

Remember:

If payback score is 100, savings = Membership Fee of \$65

If payback score is 200, savings = 2 times the Fee

If payback score is 50 savings = ½ of the Membership Fee

- Payback Score calculation**
Savings "S" / Membership Fee \$65 * 100
- Using Excel for auto-calculation of Payback Score**

Calculating the Payback Score

Annual Savings	Membership Fee	Annual Savings / Membership Fee (A/B)	Payback Score (C*100)
A	B	C	D
65	65	1	100
130	65	2	200
463.45	65	7.13	713

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ANALYTICS AT WORK

The case of Costco: Findings from analytics

The products that offered high payback scores to justify the membership fee of \$65

- Pet food: Payback Score 713
 - Price of a 38 pounds of a certain brand of dog food at Costco = \$60 vis-à-vis \$111 at any other retailer
 - Savings per purchase = \$111 - \$65 = \$46
 - Purchases required to cover the \$65 annual fee = 2 (2*\$46 = \$92)
- Diapers and baby wipes: Payback Score 373
 - Example: The Huggies brand sells for \$50, in comparison to about \$70 at other retailers
 - How many purchases by a customer would offset the Membership Fee?
 - Can you calculate the annual savings on the basis of Payback Score of 373?
- Baby formula: Payback Score 240
- Starbucks' Pike Place roast coffee beans: Payback Score 166
- Cheese: Payback Score 102
- Toilet papers: Payback Score 67.

MSN; *Business Insider*; Costco shoppers say these items basically make a membership pay for itself. We did the math; <https://www.msn.com/en-us/money/companies/costco-shoppers-say-these-items-basically-make-a-membership-pay-for-itself-we-did-the-math/ss-AA1TzJA7?ocid=hpsmn&cvid=69736122134a4db9a1275a7924269190&ei=63>
New York Post: <https://nypost.com/2026/01/13/lifestyle/costco-items-that-make-the-65-annual-membership-worthwhile/>

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ANALYTICS AT WORK

The case of Costco: Are the findings from \$65 Membership Fee research absolute?

Other business analytics for consideration

- Previous report: Costco shoppers are spending money with each minute they spend in the store.
- Data on spending habits once revealed that from the moment someone walks into a Costco, they're set to spend about \$1.75 per minute.
- An average person spends about 37 minutes wandering around Costco, which will bring the shopper to spend about \$64.75

Question: Which fields of business analytics has been covered by the Costco example?

MSN; *Business Insider*; Costco shoppers say these items basically make a membership pay for itself. We did the math; <https://www.msn.com/en-us/money/companies/costco-shoppers-say-these-items-basically-make-a-membership-pay-for-itself-we-did-the-math/ss-AA1TZjA7?ocid=hqmsn&cvid=69736122134a4db9a1275a7924269190&ei=63>
 New York Post: <https://nypost.com/2026/01/13/lifestyle/costco-items-that-make-the-65-annual-membership-worthwhile/>

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MORE ABOUT DESCRIPTIVE ANALYTICS

Examples of Descriptive Analytics

- Web traffic reports: e.g., which media channels drive the most traffic to the product page of your company's website
- Financial statements: include the balance sheet, income statement, cash flow statement
- Trends: e.g., Netflix is heavily data-driven and gathers data on users' in-platform behavior which is used to identify which TV series and movies are trending
- Survey results: e.g., to identify how preference for a product is determined by age
- Progress vis-à-vis goals and objectives: e.g., how far a company has moved towards some pre-determined key performance indicators

Harvard Business School; What is descriptive analytics: 5 examples; <https://online.hbs.edu/blog/post/descriptive-analytics>

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MORE ABOUT DIAGNOSTIC ANALYTICS

Diagnostic analytics: What it is and how it is done

Diagnostic Analytics “can be viewed as a logical next step after using descriptive analytics to identify trends. Diagnostic analysis can be done manually, using an algorithm, or with statistical software (such as Microsoft Excel).”

Some basic concepts to understand in diagnostic analysis

- Hypothesis testing: The process of proving or disproving an assumption
 - Can guide and focus diagnostic analysis
 - Example: Decline in sales in my business in January 2026 occurred due to price increase
- Correlation vs. Causation
 - Two variables move together, either in the same direction (positive) or opposite directions (negative)
 - Example of possible correlation: Level of education & salary (positive); age & health (negative)
 - **Correlation does not imply causation**
- Diagnostic regression analysis

“Determine(s) the relationship between two variables (single linear regression) or three or more variables (multiple regression). The relationship is expressed by a mathematical equation that translates to the slope of a line that best fits the variables’ relationship”

Source: Harvard Business School; What is diagnostic analytics: 4 examples; <https://online.hbs.edu/blog/post/diagnostic-analytics>

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MORE ABOUT DIAGNOSTIC ANALYTICS

Examples (Source: Harvard Business School)

- The case of HelloFresh
 - An increase noticed in fish-based recipe orders
 - Attributes of customers in the data of consumers: Females, living in northeastern US
 - Question: What kind of market research can be conducted further to diagnose the trend?
- Identifying technology issues (“running diagnostics”)
 - Example is solution-based diagnostics to detect issues and conduct a scan to determine the root cause
- Improving a company’s culture
 - HR surveys on employees’ thoughts, feelings
 - Data can be analyzed to identify factors that a company may address to increase employees’ job satisfaction
 - More resources can be allocated to these factors

Source: Harvard Business School; What is diagnostic analytics: 4 examples; <https://online.hbs.edu/blog/post/diagnostic-analytics>

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ESSENTIAL READINGS

Harvard Business School Online

Page Title: Business Insights

Available at: <https://online.hbs.edu/blog/post/types-of-data-analysis>

Harvard Business School Online

Page Title: What is descriptive analytics: 5 examples

Available at: <https://online.hbs.edu/blog/post/descriptive-analytics>

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Page Title: What is diagnostic analytics: 4 examples

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LECTURE 4

- What is Predictive Analysis
- What is Prescriptive Analysis

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